

Indian Statistical Institute
End Semestral Examination: 2024-25

Course Name: M. Math, 1st year

Subject Name : Analysis of several variables

Date: 18.11.2023, Maximum Marks: 50, Duration: Three hours

You are allowed to consult lecture notes. Question no 1. and 2. are compulsory. Answer any four problems from Question 3.

1. State and prove Green's theorem for the set

$$S = \{(x, y) \in \mathbb{R}^2 : y \geq 0, 0 \leq x^2 + y^2 \leq 1\}.$$

The statement should clearly mention the orientations and the parametrizations involved. 10

2. Let $\Omega = \{(x, y, 0) \in \mathbb{R}^3 : 1 \leq x^2 + y^2 \leq 2\}$.

- (a) Construct an orientation form η for Ω such that the volume form on Ω corresponding to η is given by $d\text{vol}_\Omega = dx \wedge dy$.
- (b) Equip $\partial_M \Omega$ with the induced orientation form. Show that $\gamma : (0, \frac{\pi}{2}) \rightarrow \mathbb{R}^3$ defined by $\gamma(t) = (\cos t, \sin t, 0)$ is a negatively oriented parametrization of the set $\{(x, y, 0) \in \mathbb{R}^3 : x^2 + y^2 = 1, x > 0, y > 0\}$.
- (c) Construct an open set $V \subseteq \mathbb{R}^2$ containing $\partial_M \Omega$ and a smooth vector field ν on V such that for all $x \in \partial_M \Omega$, $\nu(x)$ equals the unique outward pointing unit vector normal to the boundary.
- (d) Suppose f is a smooth function on $\mathbb{R}^2$ such that $\Delta f(x) = 0$ for all $x \in \Omega$ and $f(y) = 0$ for all $y \in \partial_M \Omega$. Prove that $f(x) = 0$ for all $x \in \Omega$. 5 + 8 + 2 + 5 = 20

3. Attempt any four problems: 4 x 5 = 20

- (a) Consider the function $f : \mathbb{R}^n \setminus \{0\}$, $f(x) = \|x\|$. Prove that for all x, y in $\mathbb{R}^n$, we have

$$Df(x)(y) = \frac{\langle x, y \rangle}{\|x\|}.$$

- (b) Let $O(n) = \{A \in M_n(\mathbb{R}) : A^T A = I_n\}$. If $A \in O(n)$, prove that there exists an open subset U of $\mathbb{R}^{\frac{n(n-1)}{2}}$ and a smooth one-one map $\psi : U \rightarrow M_n(\mathbb{R})$ such that $\psi(U)$ is an open subset of $O(n)$ containing A and $D\psi(u)$ has full rank for all $u \in U$.

✓(c) Show that

$$-x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2} - x_4 \frac{\partial}{\partial x_3} + x_3 \frac{\partial}{\partial x_4} - x_6 \frac{\partial}{\partial x_5} + x_5 \frac{\partial}{\partial x_6}$$

is a nowhere vanishing tangent vector field on S^5 .

✓(d) Let U be an open subset of $\mathbb{R}^n$ and $\omega \in \Omega^k(U), \eta \in \Omega^l(U)$. Prove the following statements:

✓i. If ω and η are closed, then $\omega \wedge \eta$ is closed.

✓ii. If ω is exact and η is closed, then $\omega \wedge \eta$ is exact.

(e) Let $\Omega = \{(0, y, z) \in \mathbb{R}^3 : y^2 + z^2 \leq 1, y > 0, z > 0\}$. Compute

$$\int_{\Omega} (dx \wedge dy + dy \wedge dz + dz \wedge dx).$$

(f) Consider the set

$$S = \{(x, y, z, w) \in \mathbb{R}^4 : x^2 + y^2 = 1, z^2 + w^2 = 1\}.$$

Construct a non-vanishing 2-form on S and deduce the formula of the corresponding volume form.